Super-resolution microscopy

Y19 (2019) — English translation

In 2004, physicists Eric Betzig, William Moerner and Stefan Hehl were awarded the Nobel Prize for “the development of high-resolution fluorescence microscopy.” The techniques they developed made it possible to improve the resolution of modern optical microscopes by an order of magnitude, although for a long time it was believed that optical microscopy had reached its fundamental limit of resolution (the Abbe limit). This task consists of 3 parts. Part A asks you to derive an expression for the Abbe resolution limit. In Part B you will be introduced to the phenomenon of fluorescence and shown how it can be used to overcome the Abbe limit. In Part C you will be able to describe a method for improving resolution through interference.

Fig. 1. Microphotographs of proteins in a cell obtained using traditional methods (left corners of both pictures, confocal, widefield) and using fluorescent nanoscopy (right corners of both pictures, STED, GSD).

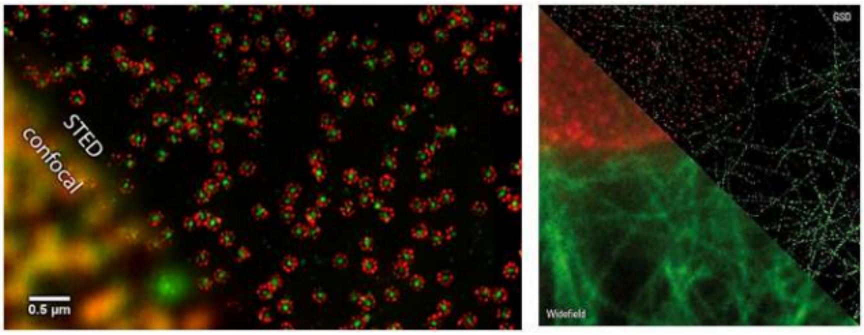

Figure 1: Figure 1.

Part A. Abbe limit

The microscope lens produces a magnified real image of the sample. The resolution of a microscope is determined primarily by the properties of the lens. When deriving the expression for the Abbe limit and further, consider an optical system consisting of a single thin lens (objective).

Fig. 2. Scheme for expressing the Abbe limit. The accuracy of the image $A'B'$ of sample AB obtained with the lens is limited due to diffraction by the lens D.

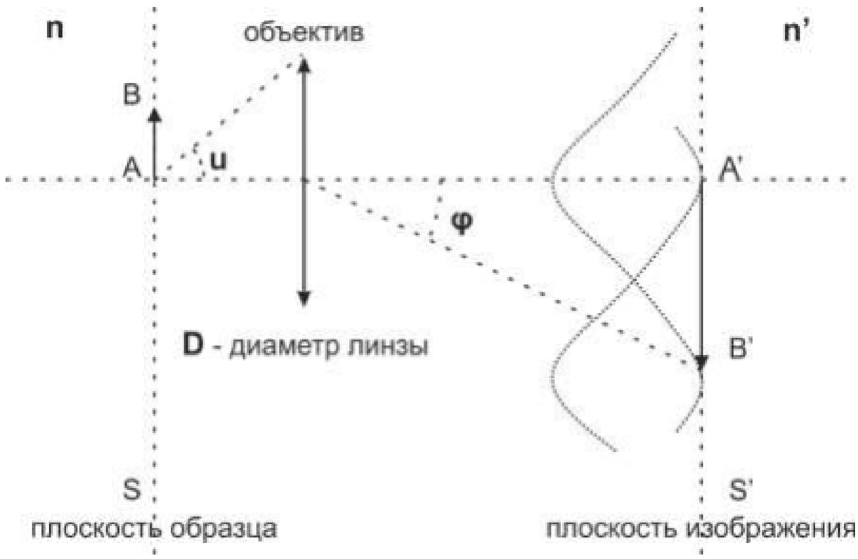

Figure 2: Figure 2.

As shown in Fig. 2., a thin lens with a diameter D is placed between two media with different refractive indices: n – on the left, on the sample side, and n' – on the right, on the image side. The resolution limit is the distance $\Delta_0 = AB$ between two such point sources A and B such that the maximum image intensity B (point B') coincides with the minimum image intensity A (point A'). In vacuum, the direction to the minimum intensity of the image A' of the source A is determined from the Fraunhofer diffraction equations on the lens $D \sin \varphi = 1.22\lambda$. When calculating, do not forget to take into account the corrections associated with the presence of matter to the left and right of the lens.

A1	Express the resolution limit $\Delta_0 = AB$ in terms of λ (the wavelength of light in vacuum), D, n, n' and $\sin u$ (the angle u is marked in Fig. 2). When drawing, use paraxial approximation.	**0.70**

A2	Initially, the system is in vacuum $n = n' = 1$. The experimenter can fill the space to the left or right of the lens (Fig. 2) with oil. Where should oil be added to improve resolution? a) from the sample side; b) from the image side	**0.30**

Part B: Understanding Fluorescence

Fluorescent molecules and substances are able to absorb photons of a certain wavelength λ_1, and then emit photons with a longer wavelength $\lambda_2 > \lambda_1$ for a short time (see Fig. 3). The probability of a molecule absorbing a photon per unit time is proportional to the intensity of the radiation incident on it I_1 and is equal to $\sigma_1 I_1$. Energy in this process goes into the external environment

Fig. 3. A fluorescent molecule in the S state can absorb a λ_1 photon, transition to the S^* state, and then emit a λ_2 photon.

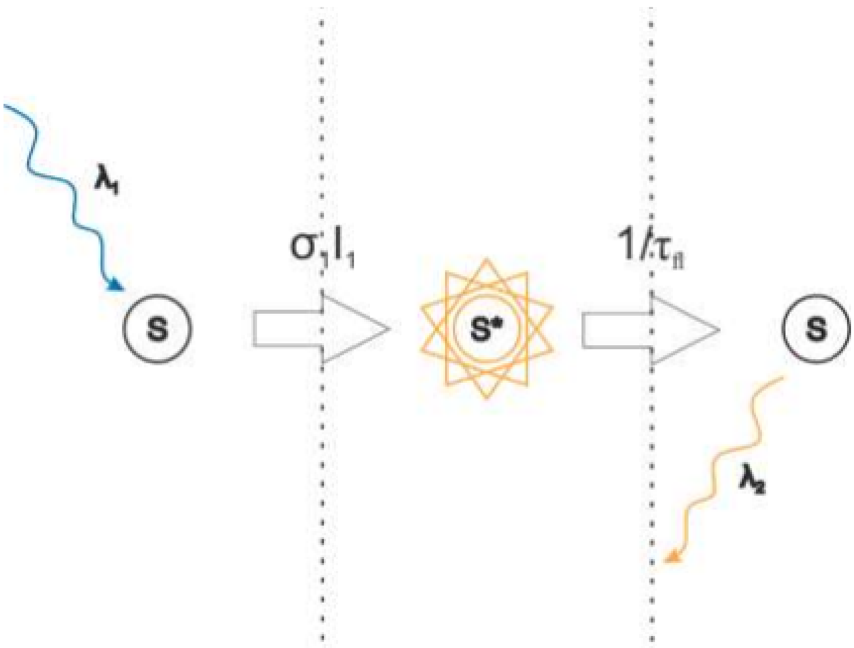

Figure 3: Figure 3.

Fluorescence obeys the laws of quantum mechanics and can be simplistically described by the following diagram of quantum states (Fig. 4).

Fig. 4. A fluorescent molecule has three states S, S^* and T and transitions between them by absorbing or emitting photons.

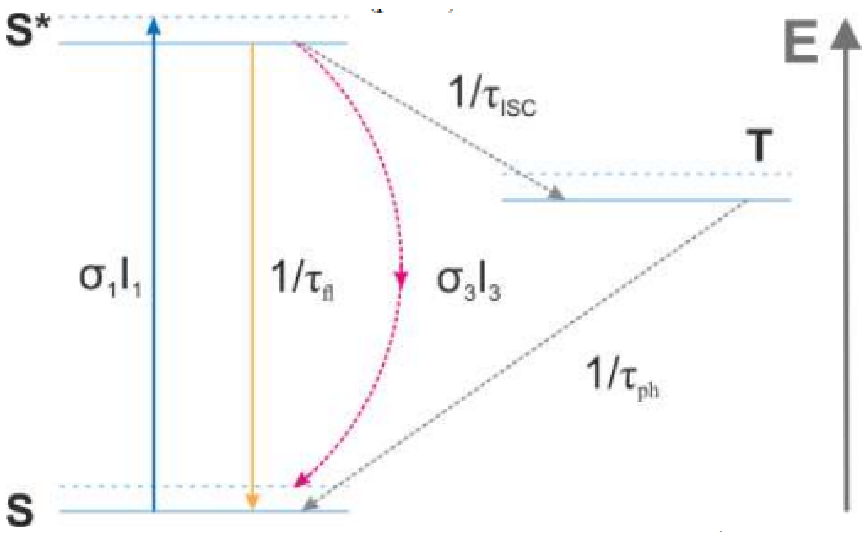

Figure 4: Figure 4.

To begin with, we will assume that the fluorescent molecule has 2 states: ground S and excited S^* (transitions to the third T state are very rare). When a photon λ_1 is absorbed, the molecule moves from S to S^*. With probability $1/\tau_{fl}$ per unit time, a molecule in state S^* emits a photon λ_2 and returns to S.

B1	Let the ensemble of $N_0 \gg 1$, initially in the S state, begin to fall on radiation with intensity I_1 and wavelength λ_1. On average, how many molecules N^* will be in the state S^* when the system reaches dynamic equilibrium?	**0.50**

B2	Plot the dependence $N^*(I_1)$ obtained in step B1 and obtain an approximation for small I_1.	**0.50**

In the future, use the approximation obtained in step B2.

Let us now consider the structure of a fluorescence microscope (Fig. 5). Radiation with wavelength λ_1 is focused by the lens in the sample plane. The sample, having absorbed photons λ_1, emits photons λ_2. The emitted photons λ_2 pass back through the lens and form an image. A special mirror, reflective for λ_1 but transparent for λ_2, is used to separate the two types of photons by wavelength.

Fig. 5. Diagram of a fluorescence microscope. The λ_1 radiation is reflected from the mirror and focused onto the sample by the lens. The emitted radiation λ_2 passes through the lens and mirror and forms an image.

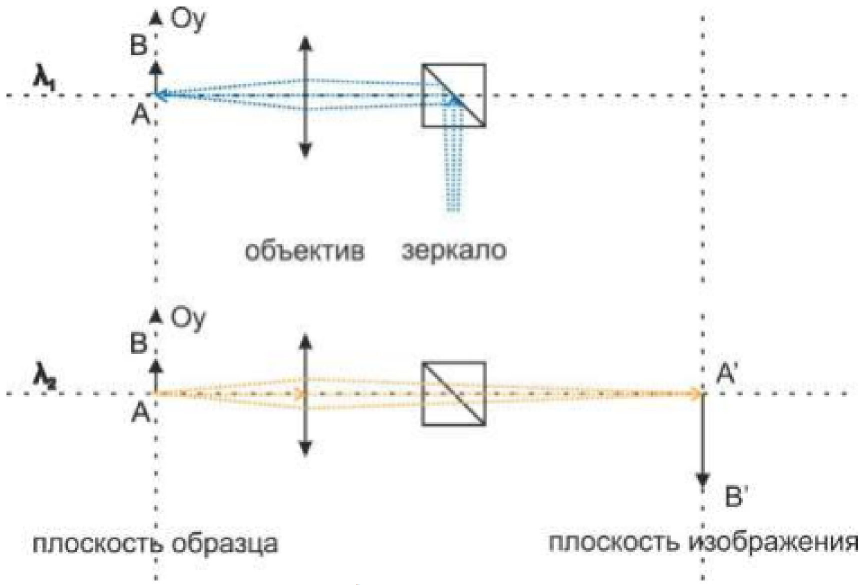

Figure 5: Figure 5.

Resolution in fluorescence microscopy is determined by the intensity distribution of the radiation λ_1 incident on the sample. Due to diffraction, its intensity in the sample plane is distributed according to the law

$$I_1(y) = I_1(0) \exp\left(-\frac{y^2}{\Delta_0^2}\right), \tag{1}$$

where Δ_0 is the Abbe resolution limit obtained in part A.

STED (STimulated Emission Depletion)

Consider the phenomenon of stimulated emission of radiation (STED): if a photon $\lambda_3 > \lambda_2, \lambda_1$ is incident on a molecule in the S^* state, the molecule can return to the S state by emitting a $\lambda_3' = \lambda_3$ photon. The probability of such a transition per unit time is $\sigma_3 I_3$ (Fig. 4). λ_3' photons emitted during stimulated emission of radiation (STED) do not participate in image formation. In those areas of the sample where such transitions occur, the number of molecules in the S^* state decreases, and hence the number of emitted λ_2 photons decreases. If the maximum radiation intensity λ_1 on the sample coincides with the minimum radiation intensity λ_3 on the sample, then this will narrow the region on it where molecules can emit photons λ_2. The size of this region was initially limited by the resolution limit Δ_0 (see formula (1)), so such an improvement would, in fact, overcome the Abbe limit. In practice, this can be done as follows. The radiation beam λ_3 is passed through a diaphragm with a diameter d. A phase plate of a certain radius is installed in the center of this diaphragm. This plate adds the π phase to the rays passing through it.

B3	At what plate radius r_0 is the minimum intensity λ_3 on the sample equal to zero?	**0.50**

In practice, phase plates with a more complex profile are often used, which give the intensity distribution λ_3 on the sample

$$I_3(y) = I_{3,\max} \cdot \left(\frac{1}{2} - \frac{1}{2} \cos \frac{\pi y}{\Delta_0} \right).$$

Such a distribution also has a characteristic width Δ_0. Since we are talking about overcoming the Abbe limit, the approximation $y \ll \Delta_0$ is appropriate.

B4	Rearrange the formula above using the $y \ll \Delta_0$ approximation.	**0.50**

There are two methods for using the STED method: cwSTED (continuous wave STED, STED with constant irradiation) and ptgSTED (pulsed time gated STED, STED with pulsed excitation and recording).

Continuous exposure STED (cwSTED)

B5 Let the sample be irradiated simultaneously by two lasers I_1 (intensity distribution (1)) and I_3 (intensity distribution from point B4). The fraction N^*/N_0 of molecules in the S^* state can be written as **1.00**

$$\frac{N^*}{N_0} = A \exp\left(-\frac{y^2}{\Delta^2_{\text{cwSTED}}} \right).$$

In practice the power $I_{3,\max}$ is large, and $\sigma_3 I_{3,\max} \gg \sigma_1 I_1(0)$. Find the parameters A and Δ_{cwSTED}. *Hint:* use the approximations $\frac{1}{1+x} \approx 1 - x \approx e^{-x}$ for $x \ll 1$.

Note that the value Δ_{cwSTED} is the resolution limit, which is defined in the same way as the Abbe resolution limit Δ_0 was previously defined.

B6	Draw the qualitative dependences $I_1(y)$, $I_3(y)$ and $N^*(y)/N_0$ on one graph.	**0.50**

B7	Determine at what $I_{3,\max}$ a tenfold gain in resolution is achieved, i.e. $\Delta_{\text{cwSTED}} = \Delta_0/10$.	**0.50**

Pulsed excitation and shooting STED (pgtSTED)

B8 Let the sample now be irradiated for time $\tau_1 \ll \tau_{fl}$ by laser I_1 (intensity distribution (1)), and then for time τ_3 by laser I_3 (intensity distribution from point B4). The fraction N^*/N_0 of molecules in the S^* state can be written as **1.00**

$$\frac{N^*}{N_0} = A \exp\left(-\frac{y^2}{\Delta^2_{\text{pgtSTED}}}\right).$$

In practice the power $I_{3,\text{max}}$ is large, and $\sigma_3 I_{3,\text{max}} \gg \sigma_1 I_1(0)$. Find the parameters A and Δ_{pgtSTED}.

Ground state depletion (GSD)

In addition to the S and S^* states, the fluorescent molecule also has a third state T. The transition $S^* \to T$ has a characteristic time τ_{ISC}, the transition $T \to S$ has a characteristic time τ_{ph}, and $\tau_{ph} \gg \tau_{ISC} \gg \tau_{fl}, 1/(\sigma_1 I_1)$. (Assume that the molecule does not return from state T to state S during the experiment.) Let laser radiation λ_1 fall on the sample during time τ_1' with distribution

$$I_1' = I'_{1,\text{max}} \left(\frac{1}{2} - \frac{1}{2} \cos \frac{\pi y}{\Delta_0}\right).$$

Then, for some time τ_1 fluorescence is excited by the same laser λ_1, but with intensity distribution (1). Moreover, $\tau_{fl} \ll \tau_1 \ll \tau_{ph}$. Here, as before, the approximations of points B2 and B4 are valid.

B9 The fraction N^*/N_0 of molecules in the S^* state can be written as **2.00**

$$\frac{N^*}{N_0} = A \exp\left(-\frac{y^2}{\Delta^2_{\text{GSD}}}\right).$$

Find A and Δ_{GSD}.

Part C. Interference 4π-microscopy and improvement of axial resolution

Part B examined techniques to improve the resolution of a fluorescence microscope in the sample plane. The problem of resolution along the optical axis is more acute - it is initially worse. To solve this problem, interference microscopy is used.

Fig. 6. An interference circuit that allows you to increase the resolution of the microscope along the optical axis. The signals received by the two lenses are combined using 50:50 reflective elements and phase plates.

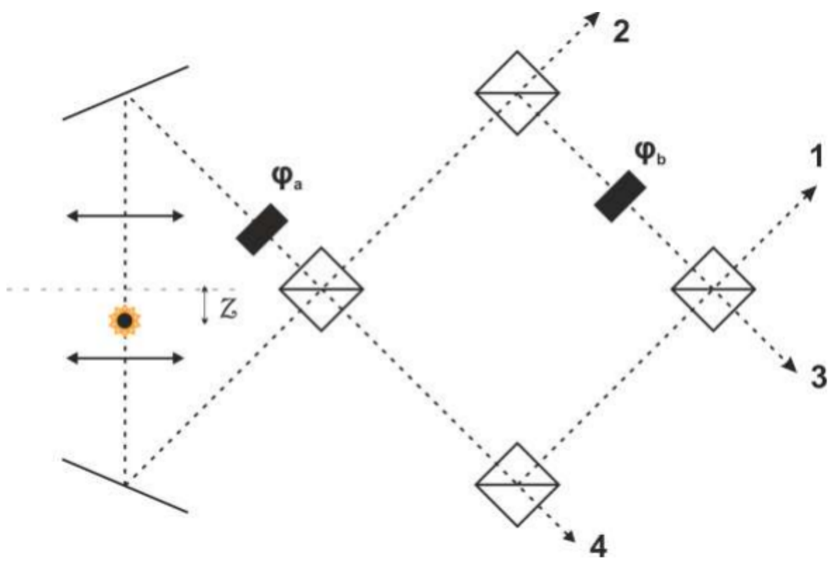

Figure 6: Figure 6.

In this setup, the image is obtained by two lenses adjusted to the same sample plane. The signals received by the lenses are divided into prisms 50 : 50, and the reflected beam gains an additional phase $\pi/2$. The optical path also contains two phase plates $\varphi_a = 3\pi/2$, $\varphi_b = 3\pi/2$. With the help of these optical elements, the signal emitted by the sample is divided into several beams with different phases and interferes, being collected at the detectors 1, 2, 3 and 4. We will assume that there is a single point source in the field of view, shifted from the plane to which the lenses are adjusted by $z < \lambda$.

C1	Let the signal intensities collected by each lens be equal to I_0. Find the intensities of the beams resulting from interference at detectors 1, 2, 3 and 4.	**2.00**

C2	Draw graphs of the dependencies found in C1.	**0.50**

C3	Express z through the relations I_1/I_3 and I_2/I_4, indicate at what shift $z \to z + \Delta z$ these relations remain unchanged?	**0.50**

Source: `https://pho.rs/p/3047`